You said:

```
(.venv)
busabhavaram@BhavaRam:/mnt/c/Users/busab/SEM5/CN/CN_PRO
JECT2$ tshark -r
captures/capture_09_chrome_example_com.pcapng -Y
"tls.handshake.type == 1" -T fields -e ip.dst -e
tls.handshake.extensions_server_name
142.251.153.119 www.google.com
35.190.80.143   beacons.gcp.gvt2.com
142.250.118.101 google.com
```

ChatGPT said:

You said:

```
(.venv)
busabhavaram@BhavaRam:/mnt/c/Users/busab/SEM5/CN/CN_PRO
JECT2$ python src/ja3.py
captures/capture_09_chrome_example_com.pcapng
Capture file: captures/capture_09_chrome_example_com.pcapng
Total packets: 107

ClientHello #1
  Source:       10.7.22.45
  Destination:  142.251.153.119
  JA3 string:   771,4865-4866-4867-49195-49199-49196-49200-
52393-52392-49171-49172-156-157-47-53,51764-27-65281-16-5-
43-17613-13-51-45-18-0-11-10-23-35-65037,4588-29-23-24,0
  JA3 MD5:      7f638a8c9da6234f9bfe77505b78ee23

ClientHello #2
  Source:       10.7.22.45
  Destination:  35.190.80.143
  JA3 string:   771,4865-4866-4867-49195-49199-49196-49200-
52393-52392-49171-49172-156-157-47-53,51764-45-17613-27-51-
13-5-0-18-35-65281-10-65037-16-43-23-11-41,4588-29-23-24,0
  JA3 MD5:      ed8992b47e8dae57b52cb3faee63b60e

ClientHello #3
  Source:       10.7.22.45
  Destination:  142.250.118.101
  JA3 string:   771,4865-4866-4867-49195-49199-49196-49200-
52393-52392-49171-49172-156-157-47-53,13-65281-18-51764-51-
```

65037-45-27-35-10-17613-16-0-5-23-43-11-41,4588-29-23-24,0
JA3 MD5: 872a1d3d9de411f93c3a1b7cb288fda3

ClientHello messages processed: 3

ChatGPT said:

You said:

.venv)
busabhavaram@BhavaRam:/mnt/c/Users/busab/SEM5/CN/CN_PROJECT2$ tshark -r captures/capture_09_chrome_example_com.pcapng -Y "tls.handshake.type == 1" -T fields -e ip.dst -e tls.handshake.extensions_server_name
104.18.39.21 ws.chatgpt.com
192.178.177.94 fonts.gstatic.com
172.66.147.243 example.com
104.20.23.154 example.com
3.217.218.228 win-extension.femetrics.grammarly.io
(.venv)
busabhavaram@BhavaRam:/mnt/c/Users/busab/SEM5/CN/CN_PROJECT2$ tshark -r captures/capture_09_chrome_example_com.pcapng -Y "tls.handshake.type == 1" -T fields -e ip.dst -e tls.handshake.extensions_server_name
172.66.147.243 example.com
104.20.23.154 example.com
(.venv)
busabhavaram@BhavaRam:/mnt/c/Users/busab/SEM5/CN/CN_PROJECT2$ python src/inspect_tls.py captures/capture_09_chrome_example_com.pcapng
Capture file: captures/capture_09_chrome_example_com.pcapng
Packets returned by Scapy (TCP session mode): 52

ClientHello #1
Source: 10.7.22.45
Destination: 172.66.147.243
Legacy version: 771
Cipher suites: [39578, 4865, 4866, 4867, 49195, 49199, 49196, 49200, 52393, 52392, 49171, 49172, 156, 157, 47, 53]
Extension IDs: [2570, 51, 16, 43, 35, 18, 23, 5, 11, 0, 45, 17613, 13, 51764, 65037, 65281, 10, 27, 60138]
Supported groups: [39578, 4588, 29, 23, 24]
EC point formats: [0]

ServerHello #1
Source: 172.66.147.243
Destination: 10.7.22.45
Legacy version: 771
Selected cipher: 4865
Extension IDs: [51, 43]

ClientHello #2
Source: 10.7.22.45
Destination: 104.20.23.154
Legacy version: 771
Cipher suites: [4866, 4865, 49196, 49195, 49200, 49199, 49188, 49187, 49192, 49191, 49162, 49161, 49172, 49171, 157, 156, 61, 60, 53, 47]
Extension IDs: [0, 43, 13, 35, 10, 11, 51, 49, 23, 65281, 45, 41]
Supported groups: [29, 23, 24]
EC point formats: [0]

ServerHello #2
Source: 104.20.23.154
Destination: 10.7.22.45
Legacy version: 771
Selected cipher: 4866
Extension IDs: [41, 51, 43]

--- Summary ---
ClientHello messages parsed: 2
ServerHello messages parsed: 2

see i have captured again using chrome by deleting previous files of chrome and then filetered example.com only for our purpose as you can see here

ChatGPT said:

You said:

(.venv) busabhavaram@BhavaRam:/mnt/c/Users/busab/SEM5/CN/CN_PROJECT2$ python src/ja3.py captures/capture_09_chrome_example_com.pcapng
Capture file: captures/capture_09_chrome_example_com.pcapng
Total packets: 52

ClientHello #1
Source: 10.7.22.45
Destination: 172.66.147.243
JA3 string: 771,4865-4866-4867-49195-49199-49196-49200-52393-52392-49171-49172-156-157-47-53,51-16-43-35-18-23-5-11-0-45-17613-13-51764-65037-65281-10-27,4588-29-23-24,0
JA3 MD5: 7fad526f9d93d143d9390fbfa3be30bb

ClientHello #2
Source: 10.7.22.45
Destination: 104.20.23.154
JA3 string: 771,4866-4865-49196-49195-49200-49199-49188-49187-49192-49191-49162-49161-49172-49171-157-156-61-60-53-47,0-43-13-35-10-11-51-49-23-65281-45-41,29-23-24,0
JA3 MD5: 6ad69cecca55b3321a3261b1ae4aaab1

ClientHello messages processed: 2

we can observe that extension re-ordering happened and even chrome used different extensions in different requests which is it's behaviour from chrome version ...

Worked for 1m 20s >

ChatGPT said:

You said:

what to do next

ChatGPT said:

You said:

(.venv) busabhavaram@BhavaRam:/mnt/c/Users/busab/SEM5/CN/CN_PROJECT2$ curl -V
curl 8.5.0 (x86_64-pc-linux-gnu) libcurl/8.5.0 OpenSSL/3.0.13 zlib/1.3 brotli/1.1.0 zstd/1.5.5 libidn2/2.3.7 libpsl/0.21.2 (+libidn2/2.3.7) libssh/0.10.6/openssl/zlib nghttp2/1.59.0 librtmp/2.3 OpenLDAP/2.6.10
Release-Date: 2023-12-06, security patched: 8.5.0-2ubuntu10.16
Protocols: dict file ftp ftps gopher gophers http https imap imaps ldap ldaps mqtt pop3 pop3s rtmp rtsp scp sftp smb smbs smtp

smtps telnet tftp
Features: alt-svc AsynchDNS brotli GSS-API HSTS HTTP2 HTTPS-proxy IDN IPv6 Kerberos Largefile libz NTLM PSL SPNEGO SSL threadsafe TLS-SRP UnixSockets zstd
the first capture (first_https.pcap) is curl based know,have you forgot? if you forgot then tell me i will redo it?as at the end you need to prepare a report(i will share the format) and slides(will share the format)

Worked for 9s >

ChatGPT said:

You said:

let's do

ChatGPT said:

You said:

done

Worked for 1m 31s >

ChatGPT said:

You said:

done,let's go

ChatGPT said:

You said:

(.venv)
busabhavaram@BhavaRam:/mnt/c/Users/busab/SEM5/CN/CN_PROJECT2$ python src/identify.py captures/capture_09_chrome_example_com.pcapng
Capture file: captures/capture_09_chrome_example_com.pcapng
Total packets: 52

ClientHello #1
  Source: 10.7.22.45

Destination: 172.66.147.243
JA3 string: 771,4865-4866-4867-49195-49199-49196-49200-52393-52392-49171-49172-156-157-47-53,51-16-43-35-18-23-5-11-0-45-17613-13-51764-65037-65281-10-27,4588-29-23-24,0
JA3 hash: 7fad526f9d93d143d9390fbfa3be30bb
Matching database records: 1

Candidate #1
Client: Chrome
Client version: Not recorded
TLS library: Chromium TLS stack
Library version: Not recorded
Notes: First observed example.com ClientHello. GREASE values removed from JA3.

ClientHello #2
Source: 10.7.22.45
Destination: 104.20.23.154
JA3 string: 771,4866-4865-49196-49195-49200-49199-49188-49187-49192-49191-49162-49161-49172-49171-157-156-61-60-53-47,0-43-13-35-10-11-51-49-23-65281-45-41,29-23-24,0
JA3 hash: 6ad69cecca55b3321a3261b1ae4aaab1
Matching database records: 1

Candidate #1
Client: Chrome
Client version: Not recorded
TLS library: Chromium TLS stack
Library version: Not recorded
Notes: Second observed example.com ClientHello. Different cipher and extension configuration from the first.

--- Summary ---
ClientHello messages processed: 2
Messages with database matches: 2
Messages without database matches: 0

i couldn't understand what output mean

ChatGPT said:

You said:

(.venv)
busabhavaram@BhavaRam:/mnt/c/Users/busab/SEM5/CN/CN_PRO

JECT2$ python src/identify.py captures/first_https.pcap
Capture file: captures/first_https.pcap
Total packets: 16

ClientHello #1
Source: 172.23.55.87
Destination: 104.20.23.154
JA3 string: 771,4866-4867-4865-49196-49200-159-52393-52392-52394-49195-49199-158-49188-49192-107-49187-49191-103-49162-49172-57-49161-49171-51-157-156-61-60-53-47-255,0-11-10-16-22-23-49-13-43-45-51-21,29-23-30-25-24-256-257-258-259-260,0-1-2
JA3 hash: 0149f47eabf9a20d0893e2a44e5a6323
Matching database records: 2

Candidate #1
Client: curl
Client version:8.5.0
TLS library: OpenSSL
Library version: 3.0.13
Notes: Captured using curl. Uses OpenSSL 3.0.13.

Candidate #2
Client: Python Requests
Client version: Not recorded
TLS library: OpenSSL
Library version: 3.0.13
Notes: Produced the same JA3 hash as the curl experiment.

--- Summary ---
ClientHello messages processed: 1
Messages with database matches: 1
Messages without database matches: 0
i wonder why some fields were held not recorded?

ChatGPT said:

You said:

Google Chrome	**156.0.8078.80**

PROJECT2$ openssl version
OpenSSL 3.0.13 30 Jan 2024 (Library: OpenSSL 3.0.13 30 Jan 2024)
(.venv)

```
busabhavaram@BhavaRam:/mnt/c/Users/busab/SEM5/CN/CN_PRO
JECT2$ python --version
Python 3.12.3
(.venv)
busabhavaram@BhavaRam:/mnt/c/Users/busab/SEM5/CN/CN_PRO
JECT2$ python -c "import ssl;print(ssl.print(ssl.OPENSSL_VERSION)"
  File "<string>", line 1
    import ssl;print(ssl.print(ssl.OPENSSL_VERSION)
                     ^
SyntaxError: '(' was never closed
(.venv)
busabhavaram@BhavaRam:/mnt/c/Users/busab/SEM5/CN/CN_PRO
JECT2$ python -c "import ssl;print(ssl.OPENSSL_VERSION)"
OpenSSL 3.0.13 30 Jan 2024
(.venv)
busabhavaram@BhavaRam:/mnt/c/Users/busab/SEM5/CN/CN_PRO
JECT2$ python -c "import requests;print(requests.__version__)"
2.34.2
(.venv)
busabhavaram@BhavaRam:/mnt/c/Users/busab/SEM5/CN/CN_PRO
JECT2$ go version
go version go1.22.2 linux/amd64
(.venv)
busabhavaram@BhavaRam:/mnt/c/Users/busab/SEM5/CN/CN_PRO
JECT2$ gnutls-cli --version | head -n 1
gnutls-cli 3.8.3
```

ChatGPT said:

You said:

see the fingerprints.json could be improved better know, we need matching ones hash, fingerprint mechanism(ja3/ja4 if possible we include ja4 in future),tls library,tls_library version,client model name,fingerprint string,endpoint(client/server)

Worked for 1m 58s >

ChatGPT said:

You said:


```
{
  "schema_version": 1,
  "description": "Experimentally collected JA3 fingerprints",
  "match_policy": "One JA3 hash may match multiple client records. Never assume a unique client from a JA3 hash alone.",
  "fingerprints": [
    {
      "client": "curl",
      "client_version": "8.5.0",
      "tls_library": "OpenSSL",
      "tls_library_version": "3.0.13",
      "capture_files": ["captures/first_https.pcap"],
      "sample_count": 1,
      "ja3_hash": "0149f47eabf9a20d0893e2a44e5a6323",
      "ja3_string": "771,4866-4867-4865-49196-49200-159-52393-52392-52394-49195-49199-158-49188-49192-107-49187-49191-103-49162-49172-57-49161-49171-51-157-156-61-60-53-47-255,0-11-10-16-22-23-49-13-43-45-51-21,29-23-30-25-24-256-257-258-259-260,0-1-2",
      "notes": "Captured using curl. Uses OpenSSL 3.0.13."
    },
    {
      "client": "OpenSSL CLI",
      "client_version": "3.0.13",
      "tls_library": "OpenSSL",
      "tls_library_version": "3.0.13",
      "capture_files": [
        "captures/capture_02_openssl_cli_example_com.pcap",
        "captures/capture_03_openssl_repeatibilty_example_com.pcap"
      ],
      "sample_count": 4,
      "ja3_hash": "a3afc2c46ba4a7d7fbe1cfb7a3031c2f",
      "ja3_string": null,
      "notes": "Fingerprint remained consistent across repeated connections."
    },
    {
      "client": "OpenSSL CLI with ALPN",
      "client_version": "3.0.13",
      "tls_library": "OpenSSL",
      "tls_library_version": "3.0.13",
      "capture_files": ["captures/capture_08_openssl_alpn_example_com.pcap"],
```

```
    "sample_count": 1,
    "ja3_hash": "5a1edc7f170af1014fc65c994878e63c",
    "ja3_string": null,
    "notes": "ALPN-enabled handshake produced a different JA3
fingerprint."
  },
  {
    "client": "Python SSL",
    "client_version": "3.12.3",
    "tls_library": "OpenSSL",
    "tls_library_version": "3.0.13",
    "capture_files":
["captures/capture_04_python_ssl_example_com.pcap"],
    "sample_count": 1,
    "ja3_hash": "d39e1be3241d516b1f714bd47c2bc968",
    "ja3_string": null,
    "notes": "Python ssl module using OpenSSL."
  },
  {
    "client": "Python Requests",
    "client_version": "2.34.2",
    "tls_library": "OpenSSL",
    "tls_library_version": "3.0.13",
    "capture_files":
["captures/capture_05_python_requests_example_com.pcap"],
    "sample_count": 1,
    "ja3_hash": "0149f47eabf9a20d0893e2a44e5a6323",
    "ja3_string": "771,4866-4867-4865-49196-49200-159-52393-
52392-52394-49195-49199-158-49188-49192-107-49187-49191-
103-49162-49172-57-49161-49171-51-157-156-61-60-53-47-
255,0-11-10-16-22-23-49-13-43-45-51-21,29-23-30-25-24-256-
257-258-259-260,0-1-2",
    "notes": "Produced the same JA3 hash as the curl experiment."
  },
  {
    "client": "Go crypto/tls",
    "client_version": "1.22.2",
    "tls_library": "Go crypto/tls",
    "tls_library_version": "1.22.2",
    "capture_files":
["captures/capture_06_go_crypto_tls_example_com.pcap"],
    "sample_count": 1,
    "ja3_hash": "88476a293aeb6f5dc7a3630d391c2980",
    "ja3_string": "771,49195-49199-49196-49200-52393-52392-
49161-49171-49162-49172-49170-4865-4866-4867,0-5-10-11-13-
```

65281-23-18-43-51,29-23-24-25,0",
"notes": "Captured using Go crypto/tls."
},
{
"client": "GnuTLS CLI",
"client_version": "3.8.3",
"tls_library": "GnuTLS",
"tls_library_version": "3.8.3",
"capture_files":
["captures/capture_07_gnutls_cli_example_com.pcap"],
"sample_count": 1,
"ja3_hash": "75484e4f01a386b961233d4238ccac89",
"ja3_string": "771,4866-4867-4865-4868-49196-52393-49325-
49162-49195-49324-49161-49200-52392-49172-49199-49171-157-
49309-53-156-49308-47-159-52394-49311-57-158-49310-51,11-
13-22-51-65281-28-45-43-35-10-5-0-23,23-24-25-29-30-256-257-
258-259-260,0",
"notes": "Captured using GnuTLS 3.8.3."
},
{
"client": "Chrome",
"client_version": "156.0.8078.80",
"tls_library": "Chromium TLS stack",
"tls_library_version": null,
"capture_files":
["captures/capture_09_chrome_example_com.pcapng"],
"sample_count": 1,
"ja3_hash": "7fad526f9d93d143d9390fbfa3be30bb",
"ja3_string": "771,4865-4866-4867-49195-49199-49196-49200-
52393-52392-49171-49172-156-157-47-53,51-16-43-35-18-23-5-
11-0-45-17613-13-51764-65037-65281-10-27,4588-29-23-24,0",
"notes": "First observed example.com ClientHello. GREASE
values removed from JA3."
},
{
"client": "Chrome",
"client_version": "156.0.8078.80",
"tls_library": "Chromium TLS stack",
"tls_library_version": null,
"capture_files":
["captures/capture_09_chrome_example_com.pcapng"],
"sample_count": 1,
"ja3_hash": "6ad69cecca55b3321a3261b1ae4aaab1",
"ja3_string": "771,4866-4865-49196-49195-49200-49199-49188-
49187-49192-49191-49162-49161-49172-49171-157-156-61-60-

53-47,0-43-13-35-10-11-51-49-23-65281-45-41,29-23-24,0",
"notes": "Second observed example.com ClientHello. Different cipher and extension configuration from the first."
}
]
}

yes,modify this

ChatGPT said:

You said:

Pasted text(3).txt
Document

why can't you understand see how my fellow mates designed correctly
instead of pasting it i mentioned the ideas,so that you may not follow exact same names and implementation which leads plaigiarism, please make a good design and if you need any values ask me,instead of populating any unknown values or nulll

Worked for 1m 10s

ChatGPT said:

You said:

(.venv) busabhavaram@BhavaRam:/mnt/c/Users/busab/SEM5/CN/CN_PROJECT2$ python src/ja3.py captures/capture_02_openssl_cli_example_com.pcap
Capture file: captures/capture_02_openssl_cli_example_com.pcap
Total packets: 13

ClientHello #1
Source: 172.23.55.87
Destination: 104.20.23.154
JA3 string: 771,4866-4867-4865-49196-49200-159-52393-52392-52394-49195-49199-158-49188-49192-107-49187-49191-103-49162-49172-57-49161-49171-51-157-156-61-60-53-47-255,0-11-10-35-22-23-13-43-45-51,29-23-30-25-24-256-257-258-259-260,0-1-2

JA3 MD5: a3afc2c46ba4a7d7fbe1cfb7a3031c2f

ClientHello messages processed: 1
(.venv)
busabhavaram@BhavaRam:/mnt/c/Users/busab/SEM5/CN/CN_PROJECT2$ python src/ja3.py captures/capture_08_openssl_alpn_example_com.pcap
Capture file: captures/capture_08_openssl_alpn_example_com.pcap
Total packets: 12

ClientHello #1
Source: 172.23.55.87
Destination: 104.20.23.154
JA3 string: 771,4866-4867-4865-49196-49200-159-52393-52392-52394-49195-49199-158-49188-49192-107-49187-49191-103-49162-49172-57-49161-49171-51-157-156-61-60-53-47-255,0-11-10-35-16-22-23-13-43-45-51,29-23-30-25-24-256-257-258-259-260,0-1-2
JA3 MD5: 5a1edc7f170af1014fc65c994878e63c

ClientHello messages processed: 1
(.venv)
busabhavaram@BhavaRam:/mnt/c/Users/busab/SEM5/CN/CN_PROJECT2$ python src/ja3.py captures/capture_04_python_ssl_example_com.pcap
Capture file: captures/capture_04_python_ssl_example_com.pcap
Total packets: 10

ClientHello #1
Source: 172.23.55.87
Destination: 172.66.147.243
JA3 string: 771,4866-4867-4865-49196-49200-159-52393-52392-52394-49195-49199-158-49188-49192-107-49187-49191-103-49162-49172-57-49161-49171-51-157-156-61-60-53-47-255,0-11-10-35-22-23-13-43-45-51-21,29-23-30-25-24-256-257-258-259-260,0-1-2
JA3 MD5: d39e1be3241d516b1f714bd47c2bc968

ClientHello messages processed: 1

ChatGPT said:

You said:

give the fingerprints.json file now
i think problem statement asks only client identification right?

Worked for 1m 54s >

ChatGPT said:

fingerprints.json
Code

You said:

```
import argparse
import json
from pathlib import Path

from scapy.all import IP, IPv6, load_layer, sniff
from scapy.layers.tls.handshake import TLSClientHello
from scapy.sessions import TCPSession

from ja3 import calculate_ja3


# Locate the database relative to this Python file.
DATABASE_FILE = Path(__file__).resolve().parent.parent / "data" /
"fingerprints.json"


def get_endpoints(packet):
    """Return the source and destination IP addresses."""
    if IP in packet:
        return packet[IP].src, packet[IP].dst

    if IPv6 in packet:
        return packet[IPv6].src, packet[IPv6].dst

    return "Unknown", "Unknown"


def load_database(database_path):
    """Load the fingerprint records from the JSON database."""
    with database_path.open("r", encoding="utf-8") as file:
        database = json.load(file)
```

```
    records = database.get("fingerprints")

    if not isinstance(records, list):
        raise ValueError("Database must contain a 'fingerprints' list.")

    return records


def find_matches(ja3_hash, records):
    """Return every database record matching this JA3 hash."""
    return [
        record
        for record in records
        if record.get("ja3_hash", "").lower() == ja3_hash.lower()
    ]


def display_matches(matches):
    """Display all possible clients associated with a fingerprint."""
    if not matches:
        print("  Result: Unknown client (fingerprint not in database)")
        return

    print(f"  Matching database records: {len(matches)}")

    for index, record in enumerate(matches, start=1):
        print(f"\n  Candidate #{index}")
        print(f"    Client:        {record.get('client', 'Unknown')}")
        print(f"    Client version:{record.get('client_version') or ' Not
recorded'}")
        print(f"    TLS library:   {record.get('tls_library') or 'Unknown'}")
        print(f"    Library version: {record.get('tls_library_version') or
'Not recorded'}")
        print(f"    Notes:         {record.get('notes', 'None')}")


def identify_clients(capture_file, records):
    """Extract ClientHello messages and identify their JA3
fingerprints."""
    load_layer("tls")

    packets = sniff(
        offline=str(capture_file),
        session=TCPSession,
```

```
        store=True
    )

    print(f"Capture file: {capture_file}")
    print(f"Total packets: {len(packets)}")

    client_hello_count = 0
    matched_count = 0

    for packet in packets:
        if TLSClientHello not in packet:
            continue

        client_hello_count += 1
        hello = packet[TLSClientHello]

        # Reuse our existing JA3 implementation.
        ja3_string, ja3_hash = calculate_ja3(hello)

        source, destination = get_endpoints(packet)

        print(f"\nClientHello #{client_hello_count}")
        print(f"  Source:      {source}")
        print(f"  Destination: {destination}")
        print(f"  JA3 string:  {ja3_string}")
        print(f"  JA3 hash:    {ja3_hash}")

        matches = find_matches(ja3_hash, records)

        if matches:
            matched_count += 1

        display_matches(matches)

    print("\n--- Summary ---")
    print(f"ClientHello messages processed: {client_hello_count}")
    print(f"Messages with database matches: {matched_count}")
    print(f"Messages without database matches: {client_hello_count -
matched_count}")


def main():
    parser = argparse.ArgumentParser(
        description="Identify TLS clients using JA3 fingerprints."
    )
```

```
    parser.add_argument(
        "pcap",
        type=Path,
        help="Path to the input PCAP or PCAPNG file"
    )

    parser.add_argument(
        "--database",
        type=Path,
        default=DATABASE_FILE,
        help="Path to the fingerprint JSON database"
    )

    args = parser.parse_args()

    if not args.pcap.is_file():
        parser.error(f"Capture file not found: {args.pcap}")

    if not args.database.is_file():
        parser.error(f"Fingerprint database not found: {args.database}")

    try:
        records = load_database(args.database)
        identify_clients(args.pcap, records)

    except (OSError, json.JSONDecodeError, ValueError) as error:
        parser.error(str(error))


if __name__ == "__main__":
    main()
```

yeah ,change this identify.py

Worked for 1m 54s

ChatGPT said:

You said:

```
(.venv)
busabhavaram@BhavaRam:/mnt/c/Users/busab/SEM5/CN/CN_PRO
JECT2$ python src/identify.py
captures/capture_09_chrome_example_com.pcapng
```

Capture file: captures/capture_09_chrome_example_com.pcapng
Total packets: 52

ClientHello #1
Source: 10.7.22.45
Destination: 172.66.147.243
JA3 string: 771,4865-4866-4867-49195-49199-49196-49200-52393-52392-49171-49172-156-157-47-53,51-16-43-35-18-23-5-11-0-45-17613-13-51764-65037-65281-10-27,4588-29-23-24,0
JA3 hash: 7fad526f9d93d143d9390fbfa3be30bb
Matching fingerprint entries: 1

Fingerprint entry #1: JA3 / 7fad526f9d93d143d9390fbfa3be30bb

Candidate #1
Software: Google Chrome
Software version: 156.0.8078.80
TLS library: Chromium TLS stack
Samples: 1
Captures: captures/capture_09_chrome_example_com.pcapng
Notes: First observed example.com ClientHello variant. GREASE values were removed before JA3 calculation.

ClientHello #2
Source: 10.7.22.45
Destination: 104.20.23.154
JA3 string: 771,4866-4865-49196-49195-49200-49199-49188-49187-49192-49191-49162-49161-49172-49171-157-156-61-60-53-47,0-43-13-35-10-11-51-49-23-65281-45-41,29-23-24,0
JA3 hash: 6ad69cecca55b3321a3261b1ae4aaab1
Matching fingerprint entries: 1

Fingerprint entry #1: JA3 / 6ad69cecca55b3321a3261b1ae4aaab1

Candidate #1
Software: Google Chrome
Software version: 156.0.8078.80
TLS library: Chromium TLS stack
Samples: 1
Captures: captures/capture_09_chrome_example_com.pcapng
Notes: Second observed example.com ClientHello variant. Its cipher and extension configuration differs from the first variant.

```
--- Summary ---
ClientHello messages processed: 2
Messages with known fingerprints: 2
Messages with unknown fingerprints: 0
Software candidates reported: 2
(.venv)
busabhavaram@BhavaRam:/mnt/c/Users/busab/SEM5/CN/CN_PRO
JECT2$ python src/identify.py captures/first_https.pcap
Capture file: captures/first_https.pcap
Total packets: 16

ClientHello #1
  Source:      172.23.55.87
  Destination: 104.20.23.154
  JA3 string:  771,4866-4867-4865-49196-49200-159-52393-52392-
52394-49195-49199-158-49188-49192-107-49187-49191-103-
49162-49172-57-49161-49171-51-157-156-61-60-53-47-255,0-11-
10-16-22-23-49-13-43-45-51-21,29-23-30-25-24-256-257-258-
259-260,0-1-2
  JA3 hash:    0149f47eabf9a20d0893e2a44e5a6323
  Matching fingerprint entries: 1

  Fingerprint entry #1: JA3 / 0149f47eabf9a20d0893e2a44e5a6323

    Candidate #1
      Software:           curl
      Software version:   8.5.0
      TLS library:        OpenSSL
      TLS library version: 3.0.13
      Samples:            1
      Captures:           captures/first_https.pcap
      Notes:              Captured with curl using its OpenSSL-backed
TLS implementation.

    Candidate #2
      Software:           Python Requests
      Software version:   2.34.2
      TLS library:        OpenSSL
      TLS library version: 3.0.13
      Runtime:            Python 3.12.3
      Samples:            1
      Captures:
captures/capture_05_python_requests_example_com.pcap
      Notes:              Produced the same JA3 fingerprint as the curl
```

capture.

--- Summary ---
ClientHello messages processed: 1
Messages with known fingerprints: 1
Messages with unknown fingerprints: 0
Software candidates reported: 2

ChatGPT said:

You said:

i have validated all,go for next part

ChatGPT said:

You said:

but tests should be captured and converted to ja3 fingerprint and then we have to check against database know,for that we already have identify.py then why creating another file for test

ChatGPT said:

You said:

yeah,let's do without any timewaste

ChatGPT said:

You said:

yes it hadn't matched,but our objective is to identify atleast 5 distinct clients using our small database fingerprints know

ChatGPT said:

You said:

yes,let's do

Worked for 2m 10s

Analysis paused

Analyzed

Analyzed

Analyzed

ChatGPT said:

‹/›

You said:

```
(.venv)
busabhavaram@BhavaRam:/mnt/c/Users/busab/SEM5/CN/CN_PRO
JECT2$ python src/ja3.py
captures/capture_11_handcrafted_client.pcap
Capture file: captures/capture_11_handcrafted_client.pcap
Total packets: 14

ClientHello #1
  Source:       172.23.55.87
  Destination:  104.20.23.154
  JA3 string:   771,49200-49199-49196-49195-47-53,0-13-10-11-
16,29-23,0
  JA3 MD5:      9da9909a3bfc6e01cea783f4484afd2b

ClientHello messages processed: 1
(.venv)
busabhavaram@BhavaRam:/mnt/c/Users/busab/SEM5/CN/CN_PRO
JECT2$ python src/identify.py
captures/capture_11_handcrafted_client.pcap
Capture file: captures/capture_11_handcrafted_client.pcap
Total packets: 14

ClientHello #1
  Source:      172.23.55.87
  Destination: 104.20.23.154
  JA3 string:  771,49200-49199-49196-49195-47-53,0-13-10-11-
16,29-23,0
  JA3 hash:    9da9909a3bfc6e01cea783f4484afd2b
  Result: Unknown client (JA3 fingerprint not found in database)

--- Summary ---
ClientHello messages processed: 1
Messages with known fingerprints: 0
```

Messages with unknown fingerprints: 1
Software candidates reported: 0

ChatGPT said:

You said:

let's do

ChatGPT said:

You said:

hey why can't we test capture curl,python ssl,openssl,.. except chrome and find fingerprint and using identify.py whether they still are static or not

ChatGPT said:

You said:

(.venv)
busabhavaram@BhavaRam:/mnt/c/Users/busab/SEM5/CN/CN_PROJECT2$ python src/identify.py captures/repeat_curl.pcap
Capture file: captures/repeat_curl.pcap
Total packets: 93

ClientHello #1
  Source: 172.23.55.87
  Destination: 172.66.147.243
  JA3 string: 771,4866-4867-4865-49196-49200-159-52393-52392-52394-49195-49199-158-49188-49192-107-49187-49191-103-49162-49172-57-49161-49171-51-157-156-61-60-53-47-255,0-11-10-16-22-23-49-13-43-45-51-21,29-23-30-25-24-256-257-258-259-260,0-1-2
  JA3 hash: 0149f47eabf9a20d0893e2a44e5a6323
  Matching fingerprint entries: 1

  Fingerprint entry #1: JA3 / 0149f47eabf9a20d0893e2a44e5a6323

    Candidate #1
      Software: curl
      Software version: 8.5.0

TLS library: OpenSSL
TLS library version: 3.0.13
Samples: 1
Captures: captures/first_https.pcap
Notes: Captured with curl using its OpenSSL-backed TLS implementation.

Candidate #2
Software: Python Requests
Software version: 2.34.2
TLS library: OpenSSL
TLS library version: 3.0.13
Runtime: Python 3.12.3
Samples: 1
Captures: captures/capture_05_python_requests_example_com.pcap
Notes: Produced the same JA3 fingerprint as the curl capture.

ClientHello #2
Source: 172.23.55.87
Destination: 104.20.23.154
JA3 string: 771,4866-4867-4865-49196-49200-159-52393-52392-52394-49195-49199-158-49188-49192-107-49187-49191-103-49162-49172-57-49161-49171-51-157-156-61-60-53-47-255,0-11-10-16-22-23-49-13-43-45-51-21,29-23-30-25-24-256-257-258-259-260,0-1-2
JA3 hash: 0149f47eabf9a20d0893e2a44e5a6323
Matching fingerprint entries: 1

Fingerprint entry #1: JA3 / 0149f47eabf9a20d0893e2a44e5a6323

Candidate #1
Software: curl
Software version: 8.5.0
TLS library: OpenSSL
TLS library version: 3.0.13
Samples: 1
Captures: captures/first_https.pcap
Notes: Captured with curl using its OpenSSL-backed TLS implementation.

Candidate #2
Software: Python Requests
Software version: 2.34.2

TLS library: OpenSSL
TLS library version: 3.0.13
Runtime: Python 3.12.3
Samples: 1
Captures: captures/capture_05_python_requests_example_com.pcap
Notes: Produced the same JA3 fingerprint as the curl capture.

ClientHello #3
Source: 172.23.55.87
Destination: 172.66.147.243
JA3 string: 771,4866-4867-4865-49196-49200-159-52393-52392-52394-49195-49199-158-49188-49192-107-49187-49191-103-49162-49172-57-49161-49171-51-157-156-61-60-53-47-255,0-11-10-16-22-23-49-13-43-45-51-21,29-23-30-25-24-256-257-258-259-260,0-1-2
JA3 hash: 0149f47eabf9a20d0893e2a44e5a6323
Matching fingerprint entries: 1

Fingerprint entry #1: JA3 / 0149f47eabf9a20d0893e2a44e5a6323

Candidate #1
Software: curl
Software version: 8.5.0
TLS library: OpenSSL
TLS library version: 3.0.13
Samples: 1
Captures: captures/first_https.pcap
Notes: Captured with curl using its OpenSSL-backed TLS implementation.

Candidate #2
Software: Python Requests
Software version: 2.34.2
TLS library: OpenSSL
TLS library version: 3.0.13
Runtime: Python 3.12.3
Samples: 1
Captures: captures/capture_05_python_requests_example_com.pcap
Notes: Produced the same JA3 fingerprint as the curl capture.

ClientHello #4

Source: 172.23.55.87
Destination: 172.66.147.243
JA3 string: 771,4866-4867-4865-49196-49200-159-52393-52392-52394-49195-49199-158-49188-49192-107-49187-49191-103-49162-49172-57-49161-49171-51-157-156-61-60-53-47-255,0-11-10-16-22-23-49-13-43-45-51-21,29-23-30-25-24-256-257-258-259-260,0-1-2
JA3 hash: 0149f47eabf9a20d0893e2a44e5a6323
Matching fingerprint entries: 1

Fingerprint entry #1: JA3 / 0149f47eabf9a20d0893e2a44e5a6323

Candidate #1
Software: curl
Software version: 8.5.0
TLS library: OpenSSL
TLS library version: 3.0.13
Samples: 1
Captures: captures/first_https.pcap
Notes: Captured with curl using its OpenSSL-backed TLS implementation.

Candidate #2
Software: Python Requests
Software version: 2.34.2
TLS library: OpenSSL
TLS library version: 3.0.13
Runtime: Python 3.12.3
Samples: 1
Captures: captures/capture_05_python_requests_example_com.pcap
Notes: Produced the same JA3 fingerprint as the curl capture.

ClientHello #5
Source: 172.23.55.87
Destination: 172.66.147.243
JA3 string: 771,4866-4867-4865-49196-49200-159-52393-52392-52394-49195-49199-158-49188-49192-107-49187-49191-103-49162-49172-57-49161-49171-51-157-156-61-60-53-47-255,0-11-10-16-22-23-49-13-43-45-51-21,29-23-30-25-24-256-257-258-259-260,0-1-2
JA3 hash: 0149f47eabf9a20d0893e2a44e5a6323
Matching fingerprint entries: 1

```
  Fingerprint entry #1: JA3 / 0149f47eabf9a20d0893e2a44e5a6323

    Candidate #1
      Software:          curl
      Software version:  8.5.0
      TLS library:       OpenSSL
      TLS library version: 3.0.13
      Samples:           1
      Captures:          captures/first_https.pcap
      Notes:             Captured with curl using its OpenSSL-backed
TLS implementation.

    Candidate #2
      Software:          Python Requests
      Software version:  2.34.2
      TLS library:       OpenSSL
      TLS library version: 3.0.13
      Runtime:           Python 3.12.3
      Samples:           1
      Captures:
captures/capture_05_python_requests_example_com.pcap
      Notes:             Produced the same JA3 fingerprint as the curl
capture.

--- Summary ---
ClientHello messages processed: 5
Messages with known fingerprints: 5
Messages with unknown fingerprints: 0
Software candidates reported: 10
```

also,i moved this repeat_curl.pcap to experiments folder to write in report and distinguish with chrome

ChatGPT said:

You said:

```
(.venv)
busabhavaram@BhavaRam:/mnt/c/Users/busab/SEM5/CN/CN_PRO
JECT2$ python src/identify.py experiments/repeat_python_ssl.pcap
Capture file: experiments/repeat_python_ssl.pcap
Total packets: 65

ClientHello #1
  Source:      172.23.55.87
```

```
  Destination: 104.20.23.154
  JA3 string:  771,4866-4867-4865-49196-49200-159-52393-52392-
52394-49195-49199-158-49188-49192-107-49187-49191-103-
49162-49172-57-49161-49171-51-157-156-61-60-53-47-255,0-11-
10-35-22-23-13-43-45-51-21,29-23-30-25-24-256-257-258-259-
260,0-1-2
  JA3 hash:    d39e1be3241d516b1f714bd47c2bc968
  Matching fingerprint entries: 1

  Fingerprint entry #1: JA3 / d39e1be3241d516b1f714bd47c2bc968

    Candidate #1
      Software:           Python standard-library ssl client
      Software version:   3.12.3
      TLS library:        OpenSSL
      TLS library version: 3.0.13
      Samples:            1
      Captures:
captures/capture_04_python_ssl_example_com.pcap
      Notes:              Captured using Python's ssl module.

ClientHello #2
  Source:      172.23.55.87
  Destination: 172.66.147.243
  JA3 string:  771,4866-4867-4865-49196-49200-159-52393-52392-
52394-49195-49199-158-49188-49192-107-49187-49191-103-
49162-49172-57-49161-49171-51-157-156-61-60-53-47-255,0-11-
10-35-22-23-13-43-45-51-21,29-23-30-25-24-256-257-258-259-
260,0-1-2
  JA3 hash:    d39e1be3241d516b1f714bd47c2bc968
  Matching fingerprint entries: 1

  Fingerprint entry #1: JA3 / d39e1be3241d516b1f714bd47c2bc968

    Candidate #1
      Software:           Python standard-library ssl client
      Software version:   3.12.3
      TLS library:        OpenSSL
      TLS library version: 3.0.13
      Samples:            1
      Captures:
captures/capture_04_python_ssl_example_com.pcap
      Notes:              Captured using Python's ssl module.

ClientHello #3
```

```
  Source:      172.23.55.87
  Destination: 104.20.23.154
  JA3 string:  771,4866-4867-4865-49196-49200-159-52393-52392-
52394-49195-49199-158-49188-49192-107-49187-49191-103-
49162-49172-57-49161-49171-51-157-156-61-60-53-47-255,0-11-
10-35-22-23-13-43-45-51-21,29-23-30-25-24-256-257-258-259-
260,0-1-2
  JA3 hash:    d39e1be3241d516b1f714bd47c2bc968
  Matching fingerprint entries: 1

  Fingerprint entry #1: JA3 / d39e1be3241d516b1f714bd47c2bc968

    Candidate #1
      Software:          Python standard-library ssl client
      Software version:  3.12.3
      TLS library:       OpenSSL
      TLS library version: 3.0.13
      Samples:           1
      Captures:
captures/capture_04_python_ssl_example_com.pcap
      Notes:             Captured using Python's ssl module.

ClientHello #4
  Source:      172.23.55.87
  Destination: 104.20.23.154
  JA3 string:  771,4866-4867-4865-49196-49200-159-52393-52392-
52394-49195-49199-158-49188-49192-107-49187-49191-103-
49162-49172-57-49161-49171-51-157-156-61-60-53-47-255,0-11-
10-35-22-23-13-43-45-51-21,29-23-30-25-24-256-257-258-259-
260,0-1-2
  JA3 hash:    d39e1be3241d516b1f714bd47c2bc968
  Matching fingerprint entries: 1

  Fingerprint entry #1: JA3 / d39e1be3241d516b1f714bd47c2bc968

    Candidate #1
      Software:          Python standard-library ssl client
      Software version:  3.12.3
      TLS library:       OpenSSL
      TLS library version: 3.0.13
      Samples:           1
      Captures:
captures/capture_04_python_ssl_example_com.pcap
      Notes:             Captured using Python's ssl module.
```

```
ClientHello #5
  Source:      172.23.55.87
  Destination: 172.66.147.243
  JA3 string:  771,4866-4867-4865-49196-49200-159-52393-52392-
52394-49195-49199-158-49188-49192-107-49187-49191-103-
49162-49172-57-49161-49171-51-157-156-61-60-53-47-255,0-11-
10-35-22-23-13-43-45-51-21,29-23-30-25-24-256-257-258-259-
260,0-1-2
  JA3 hash:    d39e1be3241d516b1f714bd47c2bc968
  Matching fingerprint entries: 1

  Fingerprint entry #1: JA3 / d39e1be3241d516b1f714bd47c2bc968

    Candidate #1
      Software:           Python standard-library ssl client
      Software version:   3.12.3
      TLS library:        OpenSSL
      TLS library version: 3.0.13
      Samples:            1
      Captures:
captures/capture_04_python_ssl_example_com.pcap
      Notes:              Captured using Python's ssl module.

--- Summary ---
ClientHello messages processed: 5
Messages with known fingerprints: 5
Messages with unknown fingerprints: 0
Software candidates reported: 5
```

ChatGPT said:

You said:

matches perfectly, since they asked atleast five clients why can't we use standard ones like open ssl,python open ssl,python requests,curl,chrome,go,openssl custom tls,gnutls only
so that it would be lighter for me and you to do work
i will delete capture_03_openssl_repeatibility.pcap for clean looking files and open_ssl_alpn,capture_11_handcrafted.pcapng also

ChatGPT said:

You said:

perfect,
let's do repeat experiments for go,gnutls,chrome.
see i have deleted capture11_handcrafted also

Worked for 51s >

ChatGPT said:

You said:

```
package main

import (
    "crypto/tls"
    "fmt"
    "net"
    "time"
)

func main() {
    const serverName = "example.com"

    // Step 1: Establish a TCP connection.
    dialer := &net.Dialer{
        Timeout: 10 * time.Second,
    }

    tcpConn, err := dialer.Dial(
        "tcp",
        net.JoinHostPort(serverName, "443"),
    )

    if err != nil {
        fmt.Println("TCP connection failed:", err)
        return
    }
    defer tcpConn.Close()

    // Set a deadline for the TLS handshake.
    err = tcpConn.SetDeadline(time.Now().Add(10 * time.Second))
    if err != nil {
        fmt.Println("Setting deadline failed:", err)
        return
```

```
    }

    // Step 2: Configure TLS.
    config := &tls.Config{
        ServerName: serverName,
    }

    // Step 3: Wrap the existing TCP connection with TLS.
    tlsConn := tls.Client(tcpConn, config)

    // Step 4: Perform the TLS handshake.
    err = tlsConn.Handshake()
    if err != nil {
        fmt.Println("TLS handshake failed:", err)
        return
    }

    // Step 5: Display negotiated connection information.
    state := tlsConn.ConnectionState()

    fmt.Println("TLS handshake successful!")
    fmt.Printf("Negotiated TLS version: 0x%04x\n", state.Version)
    fmt.Printf(
        "Selected cipher: %s (0x%04x)\n",
        tls.CipherSuiteName(state.CipherSuite),
        state.CipherSuite,
    )
    fmt.Println("Handshake complete:", state.HandshakeComplete)

    if len(state.PeerCertificates) > 0 {
        fmt.Println(
            "Server certificate:",
            state.PeerCertificates[0].Subject,
        )
    }

    // Step 6: Close the TLS connection.
    tlsConn.Close()
}
```

this is main.go in the folder go_client in CN_PROJECT2 main folder

ChatGPT said:

You said:

all fingerprints matched,let's go to Gnutls

Worked for 23s >

ChatGPT said:

You said:

Capture file: experiments/repeat_gnutls.pcap
Total packets: 78

ClientHello #1
Source: 172.23.55.87
Destination: 172.66.147.243
JA3 string: 771,4866-4867-4865-4868-49196-52393-49325-49162-49195-49324-49161-49200-52392-49172-49199-49171-157-49309-53-156-49308-47-159-52394-49311-57-158-49310-51,11-28-22-35-45-65281-0-13-51-43-23-5-10,23-24-25-29-30-256-257-258-259-260,0
JA3 hash: 88293a967b085b0f8eec866518e0bcf9
Result: Unknown client (JA3 fingerprint not found in database)

ClientHello #2
Source: 172.23.55.87
Destination: 104.20.23.154
JA3 string: 771,4866-4867-4865-4868-49196-52393-49325-49162-49195-49324-49161-49200-52392-49172-49199-49171-157-49309-53-156-49308-47-159-52394-49311-57-158-49310-51,23-43-35-5-22-0-45-28-11-65281-51-13-10,23-24-25-29-30-256-257-258-259-260,0
JA3 hash: df2a190a11e84b8e568b49c7cd6aa1d0
Result: Unknown client (JA3 fingerprint not found in database)

ClientHello #3
Source: 172.23.55.87
Destination: 172.66.147.243
JA3 string: 771,4866-4867-4865-4868-49196-52393-49325-49162-49195-49324-49161-49200-52392-49172-49199-49171-157-49309-53-156-49308-47-159-52394-49311-57-158-49310-51,0-65281-22-13-5-43-11-45-28-23-10-35-51,23-24-25-29-30-256-257-258-259-260,0
JA3 hash: 8eb35e07eb8e9e587a0244293a2a64d4
Result: Unknown client (JA3 fingerprint not found in database)

ClientHello #4
Source: 172.23.55.87
Destination: 104.20.23.154
JA3 string: 771,4866-4867-4865-4868-49196-52393-49325-49162-49195-49324-49161-49200-52392-49172-49199-49171-157-49309-53-156-49308-47-159-52394-49311-57-158-49310-51,5-45-11-0-22-10-35-65281-28-51-43-23-13,23-24-25-29-30-256-257-258-259-260,0
JA3 hash: 62ead4963307a77a4217def49844c65b
Result: Unknown client (JA3 fingerprint not found in database)

ClientHello #5
Source: 172.23.55.87
Destination: 172.66.147.243
JA3 string: 771,4866-4867-4865-4868-49196-52393-49325-49162-49195-49324-49161-49200-52392-49172-49199-49171-157-49309-53-156-49308-47-159-52394-49311-57-158-49310-51,43-65281-0-11-22-45-35-28-5-23-51-10-13,23-24-25-29-30-256-257-258-259-260,0
JA3 hash: 42db996fdbba312ff5ba0f7936c5fcde
Result: Unknown client (JA3 fingerprint not found in database)

--- Summary ---
ClientHello messages processed: 5
Messages with known fingerprints: 0
Messages with unknown fingerprints: 5
Software candidates reported: 0

Worked for 42s >

ChatGPT said:

You said:

(.venv)
busabhavaram@BhavaRam:/mnt/c/Users/busab/SEM5/CN/CN_PROJECT2$ python src/identify.py experiments/repeat_chrome.pcapng
Capture file: experiments/repeat_chrome.pcapng
Total packets: 401

ClientHello #1
Source: 10.7.22.45
Destination: 23.2.77.217
JA3 string: 771,4866-4865-49196-49195-49200-49199-49188-

49187-49192-49191-49162-49161-49172-49171-157-156-61-60-53-47,0-43-13-35-10-11-51-49-23-65281-45-41,29-23-24,0
JA3 hash: 6ad69cecca55b3321a3261b1ae4aaab1
Matching fingerprint entries: 1

Fingerprint entry #1: JA3 / 6ad69cecca55b3321a3261b1ae4aaab1

Candidate #1
Software: Google Chrome
Software version: 156.0.8078.80
TLS library: Chromium TLS stack
Samples: 1
Captures: captures/capture_09_chrome_example_com.pcapng
Notes: Second observed example.com ClientHello variant. Its cipher and extension configuration differs from the first variant.

ClientHello #2
Source: 10.7.22.45
Destination: 34.80.0.147
JA3 string: 771,4865-4866-4867-49195-49199-49196-49200-52393-52392-49171-49172-156-157-47-53,35-45-16-65037-65281-11-23-51764-27-13-18-0-43-51-17613-5-10,4588-29-23-24,0
JA3 hash: bd7420cc5c7c8755661aa7fd9207b78a
Result: Unknown client (JA3 fingerprint not found in database)

--- Summary ---
ClientHello messages processed: 2
Messages with known fingerprints: 1
Messages with unknown fingerprints: 1
Software candidates reported: 1
i had captures using wireshark and chrome in incognito mode,since command line couldn't sciff the packets

ChatGPT said:

You said:

36	23.2.77.217	up.autodesk.com
120	34.80.0.147	beacons.gcp.gvt2.com

ChatGPT said:

You said:

```
(.venv)
busabhavaram@BhavaRam:/mnt/c/Users/busab/SEM5/CN/CN_PRO
JECT2$ tshark -r experiments/repeat_chrome.pcapng -Y
"tls.handshake.type == 1" -T fields -e frame.number -e ip.dst -e
tls.handshake.extensions_server_name
87      142.251.156.119 www.google.com
124     142.250.122.94  fonts.gstatic.com
220     104.20.23.154   example.com
258     172.66.147.243  example.com
353     142.251.156.119 www.google.com
460     142.251.156.119 www.google.com
591     142.251.156.119 www.google.com
699     142.251.156.119 www.google.com
819     142.251.156.119 www.google.com
850     142.250.122.113 android.clients.google.com
881     142.250.118.139 google.com
884     192.178.158.84  accounts.google.com
```

ChatGPT said:

You said:

```
(.venv)
busabhavaram@BhavaRam:/mnt/c/Users/busab/SEM5/CN/CN_PRO
JECT2$ python src/identify.py
experiments/repeat_chrome_example_only.pcapng
Capture file: experiments/repeat_chrome_example_only.pcapng
Total packets: 1

ClientHello #1
  Source:      10.7.22.45
  Destination: 172.66.147.243
  JA3 string:  771,4866-4865-49196-49195-49200-49199-49188-
49187-49192-49191-49162-49161-49172-49171-157-156-61-60-
53-47,0-43-13-35-10-11-51-49-23-65281-45-41,29-23-24,0
  JA3 hash:    6ad69cecca55b3321a3261b1ae4aaab1
  Matching fingerprint entries: 1

  Fingerprint entry #1: JA3 / 6ad69cecca55b3321a3261b1ae4aaab1

    Candidate #1
      Software:           Google Chrome
      Software version:   156.0.8078.80
```

TLS library: Chromium TLS stack
Samples: 1
Captures: captures/capture_09_chrome_example_com.pcapng
Notes: Second observed example.com ClientHello variant. Its cipher and extension configuration differs from the first variant.

--- Summary ---
ClientHello messages processed: 1
Messages with known fingerprints: 1
Messages with unknown fingerprints: 0
Software candidates reported: 1
2 is enough for now,because this capture is difficult

ChatGPT said:

You said:

(.venv) busabhavaram@BhavaRam:/mnt/c/Users/busab/SEM5/CN/CN_PROJECT2$ python src/ja3.py experiments/repeat_chrome.pcapng
Capture file: experiments/repeat_chrome.pcapng
Total packets: 760

ClientHello #1
Source: 10.7.22.45
Destination: 142.251.156.119
JA3 string: 771,4865-4866-4867-49195-49199-49196-49200-52393-52392-49171-49172-156-157-47-53,51764-51-45-35-65281-5-17613-10-0-11-18-27-43-13-65037-23-16-41,4588-29-23-24,0
JA3 MD5: 863ed2190fc5ba3885fc25cfd0a913c0

ClientHello #2
Source: 10.7.22.45
Destination: 142.250.122.94
JA3 string: 771,4865-4866-4867-49195-49199-49196-49200-52393-52392-49171-49172-156-157-47-53,5-13-45-11-65281-23-27-35-18-16-65037-51-17613-0-43-10-51764,4588-29-23-24,0
JA3 MD5: 4351cad3fd41ec1f5d5afb58eaa57cf2

ClientHello #3
Source: 10.7.22.45
Destination: 104.20.23.154
JA3 string: 771,4865-4866-4867-49195-49199-49196-49200-

52393-52392-49171-49172-156-157-47-53,0-11-51-51764-65037-23-16-13-18-45-35-27-65281-17613-10-5-43,4588-29-23-24,0
JA3 MD5: 6ba74ceaf2bf74ebefadece1b1a1f77d

ClientHello #4
Source: 10.7.22.45
Destination: 172.66.147.243
JA3 string: 771,4866-4865-49196-49195-49200-49199-49188-49187-49192-49191-49162-49161-49172-49171-157-156-61-60-53-47,0-43-13-35-10-11-51-49-23-65281-45-41,29-23-24,0
JA3 MD5: 6ad69cecca55b3321a3261b1ae4aaab1

ClientHello #5
Source: 10.7.22.45
Destination: 142.251.156.119
JA3 string: 771,4865-4866-4867-49195-49199-49196-49200-52393-52392-49171-49172-156-157-47-53,65281-27-45-43-65037-51764-0-5-16-23-17613-13-11-10-51-18-35-41,4588-29-23-24,0
JA3 MD5: f98c718ea3e53e43daecc36357f84bc1

ClientHello #6
Source: 10.7.22.45
Destination: 142.251.156.119
JA3 string: 771,4865-4866-4867-49195-49199-49196-49200-52393-52392-49171-49172-156-157-47-53,51764-11-43-27-16-10-35-5-0-18-65281-45-51-17613-23-13-65037,4588-29-23-24,0
JA3 MD5: 45bb00671b286e33965b2ffacc8d47ff

ClientHello #7
Source: 10.7.22.45
Destination: 142.251.156.119
JA3 string: 771,4865-4866-4867-49195-49199-49196-49200-52393-52392-49171-49172-156-157-47-53,43-65037-51-65281-27-0-23-10-18-16-35-13-51764-5-45-17613-11,4588-29-23-24,0
JA3 MD5: 79e3d961da113fd066f93a8ea07ffe56

ClientHello #8
Source: 10.7.22.45
Destination: 142.251.156.119
JA3 string: 771,4865-4866-4867-49195-49199-49196-49200-52393-52392-49171-49172-156-157-47-53,16-17613-18-65281-13-65037-5-23-51764-51-35-0-27-11-43-45-10,4588-29-23-24,0
JA3 MD5: 1f2a1768779e4f8dc0092984fa709bd1

ClientHello #9

Source: 10.7.22.45
Destination: 142.251.156.119
JA3 string: 771,4865-4866-4867-49195-49199-49196-49200-52393-52392-49171-49172-156-157-47-53,0-51764-5-43-17613-11-16-18-65037-10-51-35-13-27-65281-45-23,4588-29-23-24,0
JA3 MD5: 31042812ecfb33fe084c23b07f301e92

ClientHello #10
Source: 10.7.22.45
Destination: 142.250.122.113
JA3 string: 771,4865-4866-4867-49195-49199-49196-49200-52393-52392-49171-49172-156-157-47-53,45-23-0-18-65037-17613-13-43-51764-5-11-10-16-27-65281-35-51-41,4588-29-23-24,0
JA3 MD5: cd93975997f00c44317c61861fd7cb82

ClientHello #11
Source: 10.7.22.45
Destination: 142.250.118.139
JA3 string: 771,4865-4866-4867-49195-49199-49196-49200-52393-52392-49171-49172-156-157-47-53,16-0-27-65281-17613-13-5-10-18-51-43-65037-11-51764-45-35-23-41,4588-29-23-24,0
JA3 MD5: 56dad0eb22d89988035f38afeceddf2f

ClientHello #12
Source: 10.7.22.45
Destination: 192.178.158.84
JA3 string: 771,4865-4866-4867-49195-49199-49196-49200-52393-52392-49171-49172-156-157-47-53,65037-13-11-0-51-18-51764-23-27-16-17613-45-65281-10-43-35-5,4588-29-23-24,0
JA3 MD5: d69fb9f488a6087b64844a3ab9dfcab3

ClientHello messages processed: 12

ChatGPT said:

You said:

hey but how to sepeareate that example.com ones capture

ChatGPT said:

You said:

```
(.venv)
busabhavaram@BhavaRam:/mnt/c/Users/busab/SEM5/CN/CN_PRO
JECT2$ python src/identify.py
experiments/repeat_chrome_example_only.pcapng
Capture file: experiments/repeat_chrome_example_only.pcapng
Total packets: 70

ClientHello #1
  Source:      10.7.22.45
  Destination: 104.20.23.154
  JA3 string:  771,4865-4866-4867-49195-49199-49196-49200-
52393-52392-49171-49172-156-157-47-53,0-11-51-51764-65037-
23-16-13-18-45-35-27-65281-17613-10-5-43,4588-29-23-24,0
  JA3 hash:    6ba74ceaf2bf74ebefadece1b1a1f77d
  Result: Unknown client (JA3 fingerprint not found in database)

ClientHello #2
  Source:      10.7.22.45
  Destination: 172.66.147.243
  JA3 string:  771,4866-4865-49196-49195-49200-49199-49188-
49187-49192-49191-49162-49161-49172-49171-157-156-61-60-
53-47,0-43-13-35-10-11-51-49-23-65281-45-41,29-23-24,0
  JA3 hash:    6ad69cecca55b3321a3261b1ae4aaab1
  Matching fingerprint entries: 1

  Fingerprint entry #1: JA3 / 6ad69cecca55b3321a3261b1ae4aaab1

    Candidate #1
      Software:           Google Chrome
      Software version:   156.0.8078.80
      TLS library:        Chromium TLS stack
      Samples:            1
      Captures:
captures/capture_09_chrome_example_com.pcapng
      Notes:              Second observed example.com ClientHello
variant. Its cipher and extension configuration differs from the first
variant.

--- Summary ---
ClientHello messages processed: 2
Messages with known fingerprints: 1
Messages with unknown fingerprints: 1
Software candidates reported: 1
```

ChatGPT said:

You said:

hey those are experiments which we done against our database to check and all observations we get is to be reported in report and why they happened also
yes ,let's include JA4 now in code and fingerprints.json

Worked for 2m 15s

ChatGPT said:

You said:

why not other captures since we have all captures know

Worked for 47s

ChatGPT said:

You said:

Pasted text(4).txt
Document

Worked for 5m 59s

Analyzed
Analysis errored

ChatGPT said:

identify.py
Code

fingerprints.json
Code

You said:

why can't you give identify.py and fingerprints.json as text in the response

i can't download them

Worked for 52s >

ChatGPT said:

You said:

(.venv)
busabhavaram@BhavaRam:/mnt/c/Users/busab/SEM5/CN/CN_PROJECT2$ python src/identify.py experiments/repeat_gnutls.pcap
usage: identify.py [-h] [--database DATABASE] pcap
identify.py: error: JA3 hash/string mismatch in fingerprint record #1.

Worked for 1m 1s >

Analyzed
Analyzed
Analyzed
Analyzed
Analyzed

ChatGPT said:

‹/›

You said:

i said you to not include anything of the experiments data,because we are testing them on our small database fingerprints.json,again you have forget and included that for specific clients in the note that we done 5 independent connections and JA4 succed and not JA3 like that ,it should be experiment and we report it as our observation and finding the behaviour in our report,why can't you understand that, the fingerprints.json should have ja3,ja4 fingerprints from our capture and not from the captures in experiment captures,we use that fingerprints.json as databse to identify clients from exaperiments

Worked for 1m 51s >

ChatGPT said:

You said:

perfect, let's move ahead

Worked for 58s >

ChatGPT said:

You said:

yes those are verifed,let's move aheade

Worked for 42s >

ChatGPT said:

You said:

```
(.venv)
busabhavaram@BhavaRam:/mnt/c/Users/busab/SEM5/CN/CN_PRO
JECT2$ python src/identify.py
captures/capture_10_openssl_custom_tls.pcap
Capture file: captures/capture_10_openssl_custom_tls.pcap
Total packets: 18

ClientHello #1
  Source:      172.23.55.87
  Destination: 104.20.23.154

  JA3 string:  771,49199-255,0-11-10-35-22-23-13,29-23-30-25-
24,0-1-2
  JA3 MD5:     d1f47f694f10b1faf653048918492609
  Result: Unknown client (fingerprint not found in database)

  JA4:         t12d020700_b06afd972a5c_e7e480e5a997
  Result: Unknown client (fingerprint not found in database)

--- Summary ---
ClientHello messages processed: 1

JA3:
  Known fingerprints:   0
  Unknown fingerprints: 1
  Software candidates:  0
```

JA4:
Known fingerprints: 0
Unknown fingerprints: 1
Software candidates: 0

i will move this capture into experiment as we haven't used it in reference database(fingerprints.json)

ChatGPT said:

You said:

Project Description: Build a tool that passively captures TLS ClientHello and ServerHello messages and computes JA3/JA3S fingerprints (with JA4/JA4S as a stretch goal) to identify the client or server application/library generating the traffic. Curate a small reference database and demonstrate distinguishing real clients (browsers, curl, custom scripts) purely from their handshake fingerprint.
Tools/Technologies: : libpcap/Scapy (or raw sockets) for capture; fingerprint computation implemented per the JA3/JA4 specifications, eBPF/XDP, Packet/Flow Generators like TRex, KV-stores (Redis, Valkey).
Expected Outcome: Demonstrate the working JA3/JA3S fingerprint extractor, validated against published reference JA3 hashes for known clients; A curated database correctly identifying at least 5 distinct clients/tools by fingerprint alone. Demonstrate distinguishing, e.g., curl vs. a browser vs. a custom TLS client on the wire.
Understanding on the fingerprinting's role in security monitoring (malware C2 detection, client identification) and its limitations, including fingerprint randomization in modern browsers as an evasion technique.
is there anything we are missing apart from report and ppt

ChatGPT said:

You said:

i had restored that custom client things, what do you mean by step 2 ,is it just verifying whther still ja3s.py is consistent with code?

ChatGPT said:

You said:

```
import hashlib
import unittest
from pathlib import Path
from scapy.all import load_layer, sniff
from scapy.sessions import TCPSession
from types import SimpleNamespace
# Enable Scapy's TLS parser before importing TLS classes.
load_layer("tls")

from scapy.layers.tls.handshake import TLSClientHello,
TLSServerHello

from src.ja3 import calculate_ja3, without_grease
from src.ja3s import calculate_ja3s


PROJECT_ROOT = Path(__file__).resolve().parents[1]
# Fixed reference capture used by regression tests.
BASELINE_PCAP_FILE = (
    PROJECT_ROOT / "captures" / "first_https.pcap"
)


class FingerprintTests(unittest.TestCase):

    @classmethod
    def setUpClass(cls):
        if not BASELINE_PCAP_FILE.exists():
            raise FileNotFoundError(f"Capture not found:
{BASELINE_PCAP_FILE}")

        packets = sniff(
    offline=str(BASELINE_PCAP_FILE),
    session=TCPSession,
    store=True,
)
        cls.client_hello = next(
            (
                packet[TLSClientHello]
                for packet in packets
                if packet.haslayer(TLSClientHello)
            ),
            None,
```

```
        )

        cls.server_hello = next(
            (
                packet[TLSServerHello]
                for packet in packets
                if packet.haslayer(TLSServerHello)
            ),
            None,
        )

        if cls.client_hello is None or cls.server_hello is None:
            raise AssertionError(
                "The capture must contain both TLS handshake
messages."
            )

    def test_grease_values_are_removed(self):
        values = [0x0A0A, 4866, 0x1A1A, 47, 0xFAFA]

        result = without_grease(values)

        self.assertEqual(result, [4866, 47])

    def test_ja3_fingerprint(self):
        ja3_string, fingerprint = calculate_ja3(self.client_hello)

        self.assertEqual(
            ja3_string,
            "771,4866-4867-4865-49196-49200-159-"
            "52393-52392-52394-49195-49199-158-"
            "49188-49192-107-49187-49191-103-"
            "49162-49172-57-49161-49171-51-157-"
            "156-61-60-53-47-255,"
            "0-11-10-16-22-23-49-13-43-45-51-21,"
            "29-23-30-25-24-256-257-258-259-260,"
            "0-1-2",
        )

        self.assertEqual(
            fingerprint,
            "0149f47eabf9a20d0893e2a44e5a6323",
        )

    def test_ja3s_fingerprint(self):
```

```
        ja3s_string, fingerprint = calculate_ja3s(self.server_hello)

        self.assertEqual(ja3s_string, "771,4866,51-43")

        self.assertEqual(
            fingerprint,
            "907bf3ecef1c987c889946b737b43de8",
        )

    def test_md5_known_value(self):
        # Verify Python's MD5 operation against a standard test value.
        self.assertEqual(
            hashlib.md5(b"hello").hexdigest(),
            "5d41402abc4b2a76b9719d911017c592",
        )
    def test_published_ja3_reference_vector(self):
        # Create a small object representing the published JA3
example.
        # SimpleNamespace lets us supply the fields expected by our
function.
        hello = SimpleNamespace(
            version=769,
            ciphers=[
                47, 53, 5, 10, 49161, 49162,
                49171, 49172, 50, 56, 19, 4,
            ],
            ext=[
                SimpleNamespace(type=0),
                SimpleNamespace(type=10, groups=[23, 24, 25]),
                SimpleNamespace(type=11, ecpl=[0]),
            ],
        )

        ja3_string, fingerprint = calculate_ja3(hello)

        expected_string = (
            "769,"
            "47-53-5-10-49161-49162-49171-49172-50-56-19-4,"
            "0-10-11,"
            "23-24-25,"
            "0"
        )

        expected_fingerprint =
"ada70206e40642a3e4461f35503241d5"
```

```
        self.assertEqual(ja3_string, expected_string)
        self.assertEqual(fingerprint, expected_fingerprint)
    def test_ja3s_reference_vector(self):
        # Construct a synthetic ServerHello with known field values.
        hello = SimpleNamespace(
            version=769,
            cipher=47,
            ext=[
                SimpleNamespace(type=65281),
                SimpleNamespace(type=0),
                SimpleNamespace(type=11),
                SimpleNamespace(type=35),
                SimpleNamespace(type=5),
                SimpleNamespace(type=16),
            ],
        )

        ja3s_string, fingerprint = calculate_ja3s(hello)

        expected_string = "769,47,65281-0-11-35-5-16"
        expected_fingerprint = "836ce314215654b5b1f85f97c73e506f"

        self.assertEqual(ja3s_string, expected_string)
        self.assertEqual(fingerprint, expected_fingerprint)
    def test_ja3_ignores_grease_values(self):
        hello1 = SimpleNamespace(
            version=771,
            ciphers=[4866, 4867, 4865],
            ext=[
                SimpleNamespace(type=0),
                SimpleNamespace(type=10, groups=[29, 23]),
                SimpleNamespace(type=11, ecpl=[0]),
            ],
        )

        hello2 = SimpleNamespace(
            version=771,
            ciphers=[4866, 0x0A0A, 4867, 4865],
            ext=[
                SimpleNamespace(type=0),
                SimpleNamespace(type=0x1A1A),
                SimpleNamespace(type=10, groups=[29, 0x2A2A, 23]),
                SimpleNamespace(type=11, ecpl=[0]),
            ],
```

```
        )

        string1, hash1 = calculate_ja3(hello1)
        string2, hash2 = calculate_ja3(hello2)

        self.assertEqual(string1, string2)
        self.assertEqual(hash1, hash2)
if __name__ == "__main__":
    unittest.main()
```

this is test_fingerprints.py
is this what invoked when we call "python -m unittest discover -s tests -v"

ChatGPT said:

You said:

for what this is used for?

ChatGPT said:

You said:

just delete that test_fingerprints.py and check_reasssembly.py as i need to explain it again in presentation,we had already marked/noted fingerprints in fingerprints.json know it's enough

ChatGPT said:

You said:

they asked "Demonstrate distinguishing, e.g., curl vs. a browser vs. a custom TLS client on the wire."
so why can't we use our experiment files to show them using identify.py
put this tests as Python files to run immideatly

Worked for 9s >

ChatGPT said: